NOCTUO

PRODUCT CARD

ULTRAPLA

exceptionally easy to print | ready to use on almost any type of FDM printer | high impact strenght | good mechanical strenght | low shrinkage

1. GENERAL INFORMATION ABOUT THE PRODUCT

UltraPLA connects PLA easiness with ABS hardiness. This material is recommended for most precise prints when decent mechanical endurance while maintaining high dimensional tolerances is needed. You can get great prints in a very short time thanks to even less shrinkage than unmodified PLA.

Main UltraPLA features:

- very easy to print,
- ready to use in almost every FDM 3D Printer,
- higher impact strength,
- minimal shrinkage,
- matte surface.

2. TECHNICAL PARAMETERS

CHARACTERISTICS	TEST METHOD	TEST CONDITIONS	IU	VALUE
	ISO			
PHYSICAL				
Density	1183	-	g/cm^3	1.30
MECHANICAL				
Elastic modulus at tension	527	-	MPa	2600
Yield strength	527-1,-2	50mm/min	MPa	47
Elongation at break	527-1,-2	50mm/min	%	19
Bending stress	178	2mm/min	MPa	64
Elastic modulus at bending	178	2mm/min	MPa	2650
Notched Charpy impact strength	179-1	1eA	kJ/m^2	7

Notched Charpy Impact strength (-30°C)	179-1	1eU	kJ/m^2	n.p.
THERMAL				
Vicat softening point	306	50N	°C	60
Melt temperature	3146-C	-	°C	150-170

Tests have been done in 23°C, if it's not marked differently.

3. RECOMMENDATION OF PRINTING

UltraPLA does not require drying before using. Satisfying results could be achieved by using standard settings for PLA in available slicers and printers.

Cooling during printing is recommended.

Recommended parameters of printing:

Hotend temperature	190 - 240 °C
Bed temperature	50 - 60 °C
Print speed	< 300 mm/s

4. SAFETY NOTES

Exhaust fan is recommended.
Air filters in printer is recommended.
UltraPLA needs to be used only in well ventilated conditions.
Inhaling fumes generated during the printing must be avoided.

Generating fumes during the printing depends mainly on printing temperature. In case of visibly raising emission level, the printing needs to end. Check the hotend temperature and efficiency of the control system before using it next time.

In proper using conditions, the product doesn't endanger health.

It's forbidden to set fire or exceed decomposition temperature!

Decomposition of PLA is typically over 250 °C and it should be strictly avoided.
Main ingredient of decomposition is lactide.

Detailed safety information available in SDS.